

FUNDEDSHIFT PROP FIRM

TECHNICAL ARCHITECTURE & SYSTEMS MANUAL

Production Platform v2.0 - ECN Execution Engine & Dynamic Risk Shield

Stack:

Engine:

1. Executive Overview & Multi-Tier Topology

FundedShift is an institutional proprietary trading firm platform built to evaluate trading talent through structured capital challenges, enforce automated risk rules, and disburse bi-weekly payouts.

TIER 1: PRESENTATION & CLIENT WORKSPACE (React 19 + TypeScript + Vite)

- Trader Web Terminal: Live TradingView candlestick charts, 200ms ECN market watch, lot calculator.
- Trader Portal: Objectives, KYC upload, official certificates, payout request desk, support ticketing.
- Admin Command Suite: Trader role management, challenge pricing, KYC review queue, payout dispatch.

TIER 2: API GATEWAY & SECURITY MIDDLEWARE (Express.js 4.21 + Node 22)

- Dual Header Authentication: Authorization Bearer JWT & x-user-id fallback for microservices.
- IP Rate Limiting: Auth (40/15min), Orders (60/min), Checkout (20/min) preventing abuse.
- Security & Sanitization: PBKDF2 with SHA-512 password hashing, XSS/CORS middleware, audit logging.

TIER 3: CORE FINANCIAL & TRADING ENGINES

- Simulated ECN Execution Engine: Sub-millisecond tick matcher, slippage, and dynamic spreads.
- Dynamic Risk Shield: 200ms intraday high-water loss computation, trailing/static drawdown monitor.
- Payout & Transition Engine: 80-90% profit split computation, automated Phase 1 -> Phase 2 transitions.

TIER 4: RELATIONAL PERSISTENCE (PostgreSQL 16 Relational Engine)

- 16 Relational Tables with foreign key cascades, unique indexes, and ACID guarantees.
- Synchronous snapshot disk mirroring to .data/propfirm_database.json for instantaneous recovery.
- Immutable audit logs tracking all administrative role updates, payouts, and compliance actions.

System Performance Targets:

Market Tick: 200ms | Order Execution: < 15ms | Risk Breach Detection: < 50ms | DB Latency: < 5ms

2. End-to-End Trader Lifecycle & Evaluation Challenges

The platform provides a 3-stage evaluation framework filtering for disciplined, profitable traders.

PHASE 1: EVAL

Step 1 Evaluation

- 8% Profit Target
- 5% Daily Loss
- 10% Max Drawdown
- Unlimited Days

PHASE 2: VERIFY

Step 2 Verification

- 5% Profit Target
- 5% Daily Loss
- 10% Max Drawdown
- Unlimited Days

PHASE 3: FUNDED

Live Funded Account

- 0% Profit Target
- 80% to 90% Split
- Bi-weekly Payouts
- First Payout at 14d

PHASE 4: SCALE

Capital Scaling Plan

- +25% Balance Boost
- Every 3 Months
- 10%+ Cum. Gain
- Up to \$2,000,000

EVALUATION TIERS & SPECIFICATIONS MATRIX

Account Size	Model	Step 1 (8%)	Step 2 (5%)	Daily Limit	Max Drawdown	Leverage	Price
\$5,000							
\$10,000							
\$25,000							
\$50,000							
\$100,000							
\$200,000							
\$10,000							

AUTOMATED ACCOUNT TRANSITION & RECOVERY LIFECYCLE

1. Target Reached: Risk engine detects balance reaching profit target. All open positions are flat.
2. Scheduled Transition: Account status is updated to PASSED. An automated background worker provisions the Step 2 Verification account within 1 to 2 hours with new login credentials.
3. Step 2 Target Reached: System marks verification passed and generates live Funded Account certificate.
4. Immediate KYC Prompt: Funded account is issued with zero profit target and up to 90% profit split eligibility.
5. Account Recovery Program: Breached accounts receive discounted reset options (-20% to -50%) without rebuy.

3. Simulated MT5 Execution Engine & Dynamic Risk Shield

The platform contains an integrated high-frequency institutional simulator mimicking MT5/cTrader execution.

CORE FINANCIAL MATHEMATICS & FORMULAS

1. Floating Unrealized PnL Formulation:

BUY: Floating PnL = (Current Bid Price - Entry Price) * Lot Size * Contract Size

SELL: Floating PnL = (Entry Price - Current Ask Price) * Lot Size * Contract Size

2. Used Margin Requirement:

Used Margin = (Lot Size * Contract Size * Entry Price) / Account Leverage

3. Live Equity & Free Margin:

Live Equity = Current Balance + Sum(Open Positions Floating PnL)

Free Margin = Max(0, Live Equity - Used Margin) | Margin Level % = (Equity / Margin) * 100

DYNAMIC RISK SHIELD & DRAWDOWN COMPUTATION

RULE 1: MAXIMUM DAILY LOSS LIMIT (5% INTRADAY)

Baseline Reset: At 00:00 UTC daily, the system records:

$\text{start_of_day_baseline} = \text{Max}(\text{start_of_day_balance}, \text{start_of_day_equity})$

Live Intraday Loss Formula = $\text{Max}(0, \text{start_of_day_baseline} - \text{Live Equity})$

Breach Threshold: If Live Equity drops below (baseline * 0.95), account breaches immediately.

Notice: Daily drawdown incorporates BOTH closed trades AND open floating losses.

RULE 2: MAXIMUM OVERALL DRAWDOWN (10% STATIC / TRAILING)

1. Static Model (Evaluation Challenges): Permanent floor at $\text{starting_balance} * 0.90$.

Example: \$100,000 account floor remains permanently fixed at \$90,000.

2. Trailing Model (Instant Funding): Floor trails highest achieved equity until starting balance.

$\text{Max Loss Violation} = (\text{Live Equity} < \text{Max Permitted Drawdown Floor})$

Emergency Flattener: Instant market closure of all positions and status set to BREACHED in < 50ms.

4. PostgreSQL 16 Schema, Financial Rails & Support Desk

In production, the platform runs on PostgreSQL 16 partitioned into 16 relational tables.

POSTGRESQL 16 RELATIONAL TABLES SCHEMA

Table Name	Primary Key	Foreign Keys	Key Columns & Constraints
users			
trading_accounts			
positions			
rule_violations			
orders & payments			
payout_requests			
kyc_submissions			
support_tickets			
audit_logs			

FINANCIAL PIPELINE, KYC VERIFICATION & SUPPORT DESK

1. Identity Verification Pipeline (KYC):

- Trader submits Passport, Driver's License, or National ID under /dashboard/security.
- Enters kyc_submissions with status = PENDING. Compliance reviews on /admin/kyc.
- Approval sets user.is_verified = true, unlocking immediate profit split withdrawal eligibility.

2. Profit Split & Payout Rails:

- Formula: Trader Payout = Profit * (Trader Split % / 100) [80% Standard, 90% Scaled].
- Rails: Crypto USDT (TRC-20, ERC-20), Direct Bank Wire, UPI (India), PayPal (within 4 hours SLA).

3. Helpdesk Ticketing & Administrative Control Suite:

- Trader Support Desk (/dashboard/support): Categorized tickets with real-time threaded replies.
- Admin Support Desk (/admin/support): Split-pane console with search, status filters, and instant reply.
- Admin Users (/admin/users): Role toggling (User <-> Admin) and account suspension with DB persistence.
- Zero Mock Fallbacks: Platform verified completely decoupled from fake starter accounts.
- Production Deployment: Dockerized container with health-check monitoring and Postgres 16.